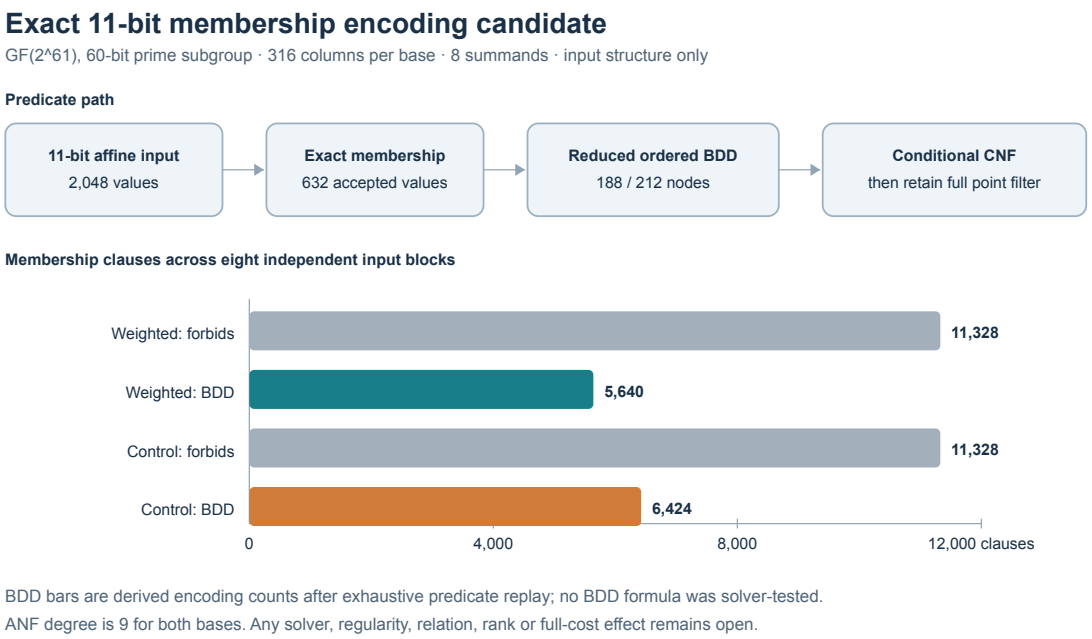
Exact membership encoding screen for two binary factor bases

Scope. Target-blind structural search on the certified ordinary-binary GF(2^61) fixture, subgroup order 576460752557211763, cofactor 4. The two archived k11 bases each contain 316 projected columns and 632 accepted coordinates in one 11-bit affine block. No solver, relation collector, degree-of-regularity measurement, rank calculation, or end-to-end comparison was run.



Exact input calculation

`analyze.py` reads the archived weighted and matched local-random bases and their XCNF manifests. It checks the algebraic normal form and both reduced ordered decision diagrams against all 2,048 Boolean inputs per predicate. The proposed BDD clause count uses one node variable, conditional equivalence clauses for each edge, and an accepted-root unit per independent block.

Quantity	Weighted	Local random
ANF terms / degree	520 / 9	516 / 9
BDD nodes, low-bit-first	188	212
BDD nodes, high-bit-first	193	216
Archived forbidden clauses, eight blocks	11,328	11,328
Proposed BDD clauses, eight blocks	5,640	6,424
Additional BDD variables, eight blocks	1,504	1,696

The archived full channel-0 formulas contain 89,122 CNF clauses and 62,655 native XOR rows for both bases. Substitution of only the membership clauses would imply 83,434 and 84,218 CNF clauses, plus the added variables shown. This is a derived description of an unbuilt encoding. Formula clause count, ANF degree, degree of regularity, solver time, and complete index-calculus cost are distinct quantities.

Provenance and next check

The exact base and manifest SHA-256 values, source digest, execution command, and full computed counts are in `result.json`. `result_initial.json` preserves a scratch metadata error in the first output: its timestamp was sampled after the calculation. The corrected output samples it before the calculation. The numerical results agree. Provenance for the frozen base and manifest records: internal; all other counts above are derived from those exact bytes.

Next, serialize the BDD encoding and independently replay its truth table and planted full-point controls. Compare both bases and both encodings on one frozen common-target panel under an admitted protocol. Charge generation, solver, model filtering, useful rank, linear algebra, target work, and matched rho before interpreting any whole-pipeline effect.

No canonical ECDLP cost graph was changed; this search establishes input predicate representation counts only.